

Network Architecture Overview

This document describes the production network architecture for the BFAI platform, including perimeter security, load balancing, and internal service segmentation. All production traffic enters through the edge firewall before reaching the load balancer tier.

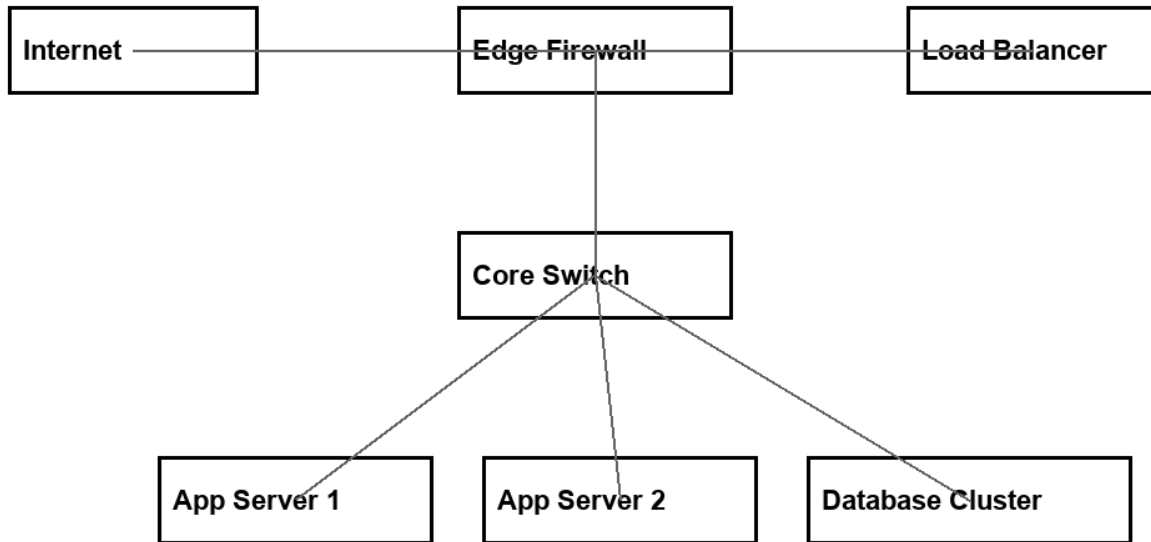


Fig. 1 - BFAI Production Network Topology (Region: ap-south-1)

The core switch routes internal traffic between the application server tier and the database cluster. Application servers are deployed across two availability zones for redundancy, and the database cluster replicates synchronously between zones.

Security Zones

The network is divided into three security zones: the public zone (internet-facing load balancers), the application zone (app servers, internal APIs), and the data zone (database cluster, object storage). Traffic between zones is restricted by firewall rules that allow only the minimum required ports.

Access to the data zone is limited to application servers and the operations team via a bastion host. All administrative access requires multi-factor authentication and is logged for audit purposes.